Optimal Fiscal Policy in a Business Cycle Model

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1 Decentralized Economy

2 Planning Problem

3 Computation Strategy

4 Findings

Roadmap

- **Decentralized economy**: environment, taxes, debt, consumer problem
- **Implementability constraint**: the key constraint for Ramsey
- **Planning problem**: Ramsey equilibrium and Ramsey allocations
- **Ramsey policies**: labor tax, capital tax, debt returns, indeterminacy
- **Computation strategy**: calibration and numerical procedure

Environment: events and histories

- In each period $t = 0, 1, \ldots$, the economy experiences one of finitely many events s_t.
- History up to t:

$$s^t \triangleq (s_0, \ldots, s_t).$$

- $\mu(s^t)$: unconditional probability of history s^t.
- Initial realization s_0 is given.
- Capital is predetermined:

$$k_t \equiv k(s^{t-1}),$$

while labor is chosen after s_t:

$$n_t \equiv n(s^t).$$

Technology and competitive pricing

Neoclassical technology:

$$y_t = F(k_t, n_t, s_t). \tag{1.1}$$

Competitive pricing:

$$r_t = F_k(k_t, n_t, s_t), \tag{1.2}$$
$$w_t = F_n(k_t, n_t, s_t), \tag{1.3}$$

where $r_t \equiv r(s^t)$ and $w_t \equiv w(s^t)$ are before-tax returns on capital and labor.

- Output uses: private consumption $c_t \equiv c(s^t)$, government consumption $g_t \equiv g(s^t)$, and saving k_{t+1}.

Preferences

Representative agent:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu(s^t) U(c_t, n_t), \tag{1.4}$$

with U satisfying standard assumptions.

- Government may levy **linear taxes** on labor and capital income.
- Lump-sum taxation is forbidden.
- Government may issue **short-term debt**.

Fiscal policy instruments

- Labor income tax rate:
$$\tau_t^n \equiv \tau^n(s^t).$$
- Capital income tax rate:
$$\tau_t^k \equiv \tau^k(s^t).$$
- Debt issued at s^t:
$$b_{t+1} \equiv b(s^t).$$
- Payoff rate at $s^{t+1} = (s^t, s_{t+1})$:
$$R_t^b \equiv R^b(s^{t+1}).$$
- Policy:
$$\pi_t = [\tau_t^n, \tau_t^k, R_t^b], \qquad \pi = [\pi_t].$$
- Initial stocks b_0 and k_0 are given.

Government budget constraint

$$g_t + R_t^b b_t = \tau_t^n w_t n_t + \tau_t^k (r_t - \delta) k_t + b_{t+1}, \tag{1.5}$$

where δ is the depreciation rate on capital.

- Government spending plus debt repayment is financed by labor taxes, capital taxes, and new debt.
- Capital tax applies to $(r_t - \delta)k_t$.

Consumer budget constraint

$$c_t + k_{t+1} + b_{t+1} \leq (1 - \tau_t^n) w_t n_t + R_t^b b_t + R_t^k k_t, \tag{1.6}$$

where the gross after-tax return on capital is

$$R_t^k \triangleq 1 + (1 - \tau_t^k)(r_t - \delta).$$

Allocation notation:

$$x_t = [c_t, n_t, k_{t+1}, b_{t+1}], \qquad x = [x_t].$$

Consumer Lagrangian

$$\mathcal{L} \triangleq \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \mu(s^t) U(c_t, n_t) + \sum_{t=0}^{\infty} \sum_{s^t} \lambda_t \left[(1-\tau_t^n) w_t n_t + R_t^b b_t + R_t^k k_t - c_t - k_{t+1} \right.$$

where

$$\lambda_t \equiv \lambda(s^t) \geq 0$$

is the per-period multiplier.

- Nonnegativity $c_t \geq 0$, $n_t \geq 0$ is implicitly required.
- By Weitzman (1973), the FOCs and TVCs below are necessary and sufficient for x to solve the consumer's problem for all π.

Consumer FOCs

$$c_t: \quad \beta^t \mu(s^t) U_{c,t} \leq \lambda_t, \quad \text{w/ equality if } c_t > 0, \tag{1.7}$$

$$n_t: \quad \beta^t \mu(s^t) U_{n,t} \leq -\lambda_t (1 - \tau_t^n) w_t, \quad \text{w/ equality if } n_t > 0, \tag{1.8}$$

$$b_{t+1}: \quad 0 = \left(\sum_{s^{t+1}} \lambda_{t+1} R_{t+1}^b - \lambda_t \right) b_{t+1}, \tag{1.9}$$

$$k_{t+1}: \quad 0 = \left(\sum_{s^{t+1}} \lambda_{t+1} R_{t+1}^k - \lambda_t \right) k_{t+1}. \tag{1.10}$$

Transversality conditions

For all s^∞:

$$\lim \lambda_t b_{t+1} = 0, \tag{1.11}$$

$$\lim \lambda_t k_{t+1} = 0, \tag{1.12}$$

where limits are taken over sequences of histories s^t contained in the infinite history s^∞.

- These TVCs rule out Ponzi schemes in debt and capital.

Implementability constraint

Using the consumer FOCs (1.7) and (1.8), and interiority from Inada conditions:

$$\boxed{\sum_{t=0}^{\infty}\sum_{s^t}\beta^t\mu(s^t)(U_{c,t}c_t+U_{n,t}n_t)=U_{c,0}(R_0^b b_0+R_0^k k_0)} \tag{1.13}$$

- This is the **implementability constraint** (IC).
- It summarizes the consumer's optimality conditions and budget constraint in terms of allocations only.
- It is the key constraint that distinguishes Ramsey from first-best.

Resource constraint

Adding the government budget constraint (1.5) to the consumer budget constraint (1.6):

$$c_t + g_t + k_{t+1} \leq y_t + (1 - \delta)k_t. \tag{1.14}$$

- Standard aggregate resource constraint.
- Debt cancels out; taxes are transfers between households and government.

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Ramsey equilibrium: definition

Ramsey Equilibrium: A policy π, an allocation rule $x(\cdot)$, and price rules $w(\cdot)$, $r(\cdot)$ such that:

(i) π maximizes (1.4) subject to (1.6), with allocations and prices given by $x(\pi)$, $w(\pi)$, $r(\pi)$;

(ii) $\forall \pi'$, $x(\pi')$ maximizes (1.4) subject to (1.6) evaluated at π' and prices $w(\pi')$, $r(\pi')$;

(iii) $\forall \pi'$, prices satisfy (1.2) and (1.3) conditional on π'.

Timing convention

- The initial capital tax rate τ_0^k is fixed.
- The initial return on debt R_0^b is fixed.
- These initial conditions matter for the period-0 allocation rules, but not for the stationary rules from period 1 onward.

Ramsey allocations: key proposition

The consumption, labor, and capital allocations $\{c_t, n_t, k_{t+1}\}$ in a Ramsey equilibrium solve the Ramsey allocation problem:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu(s^t) U(c_t, n_t)$$

subject to the implementability constraint (1.13) and the resource constraint (1.14).

- Prices and taxes are then constructed from the allocation.
- The bond allocation is constructed separately.

Constructing the bond allocation

Multiply the consumer budget constraint by λ_t, sum over all periods and states following $s^{t'}$:

$$\sum_{t=t'+1}^{\infty} \sum_{s^t} \beta^{t-t'} \mu(s^t|s^{t'})(U_{c,t}c_t + U_{n,t}n_t) = U_{c,t'}(k_{t'} + b_{t'}).$$

Hence:

$$b_{t'} = \sum_{t=t'+1}^{\infty} \sum_{s^t} \frac{1}{U_{c,t'}} \beta^{t-t'} \mu(s^t|s^{t'})(U_{c,t}c_t + U_{n,t}n_t) - k_{t'}. \tag{2.1}$$

- Given the Ramsey allocation, this pins down the debt allocation.

The modified objective

Incorporate the implementability constraint into the maximand:

$$W(c_t, n_t, \eta) \triangleq U(c_t, n_t) + \eta(U_{c,t}c_t + U_{n,t}n_t), \tag{2.2}$$

where η is the multiplier on the IC (1.13). The Ramsey allocation problem becomes:

$$\max \left\{ \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \mu(s^t) W(c_t, n_t, \eta) - \eta U_{c,0}(R_{k,0}k_0 + R_{b,0}b_0) \right\}$$

subject to the resource constraint (1.14).

Ramsey Lagrangian

$$\mathcal{L} \triangleq \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \mu(s^t) W(c_t, n_t, \eta) + \sum_{t=0}^{\infty} \sum_{s^t} \theta_t [y_t + (1-\delta)k_t - c_t - g_t - k_{t+1}] - \eta U_c,$$

where

$$\theta_t \equiv \theta(s^t) \geq 0$$

is the per-period multiplier on the resource constraint.

FOCs for $t \geq 1$

$$
\begin{aligned}
c_t: \quad & W_{c,t} = \theta_t, \\
n_t: \quad & W_{n,t} = -\theta_t F_{n,t}, \\
k_{t+1}: \quad & 0 = \sum_{s^{t+1}} \beta \mu(s^{t+1}|s^t) \theta_{t+1}(1 - \delta + F_{k,t+1}) - \theta_t.
\end{aligned}
$$

Equivalently:

$$
-\frac{W_{l,t}}{W_{c,t}} = F_{n,t}, \tag{2.3}
$$

$$
W_{c,t} - \sum_{s^{t+1}} \beta \mu(s^{t+1}|s^t) W_{c,t+1}(1 - \delta + F_{k,t+1}) = 0. \tag{2.4}
$$

FOCs for $t = 0$

$$-\frac{W_{n,0} - \eta[U_{cn,0}(R_0^k k_0 + R_0^b b_0) + U_{c,0}(1 - \tau_0^k)F_{kl,0}]}{W_{c,0} - \eta U_{cc}(R_0^k k_0 + R_0^b b_0)} = F_{n,0}, \tag{2.5}$$

$$W_{c,0} - \eta U_{cc}(R_0^k k_0 + R_0^b b_0) - \sum_{s^1} \beta \mu(s^1|s^0) W_{c,1}(1 - \delta + F_{k,1}) = 0. \tag{2.6}$$

Stationarity from period 1 onward

If the stochastic process on s is Markov:

- (2.3) and (2.4) imply that allocations from period 1 onward are described by time-invariant allocation rules:

$$c(k, s; \eta), \quad n(k, s; \eta), \quad k_+(k, s; \eta), \quad b(k, s; \eta).$$

- By (2.5) and (2.6), the period-0 allocation rules differ from the stationary rules.

From allocations to prices

Given a Ramsey allocation (c, n, k, b):

- Prices r and w are obtained from marginal products:

$$r_t = F_k(k_t, n_t, s_t), \qquad w_t = F_n(k_t, n_t, s_t).$$

- Combine with consumer FOCs to get the optimal consumption-labor choice:

$$-\frac{U_{n,t}}{U_{c,t}} = (1 - \tau_t^n) F_{n,t}, \tag{2.7}$$

  and two Euler equations:

$$U_{c,t} = \sum_{s^{t+1}} \beta \mu(s^{t+1}|s^t) U_{c,t+1} R_{t+1}^k, \tag{2.8}$$

$$U_{c,t} = \sum_{s^{t+1}} \beta \mu(s^{t+1}|s^t) U_{c,t+1} R_{t+1}^b. \tag{2.9}$$

Labor tax and capital tax

Labor tax is determined from (2.7):

$$-\frac{U_{n,t}}{U_{c,t}} = (1-\tau_t^n)F_{n,t}.$$

Capital income tax rate can be written explicitly:

$$\tau_t^k = 1 - \frac{R_t^k - 1}{F_{k,t} - \delta}.$$

- But R^b and τ^k are **not** individually pinned down.

Indeterminacy proposition

If R^b and τ^k satisfy (1.6), (2.8), and (2.9), then so do any $\bar{R}^b$ and $\bar{\tau}^k$ satisfying

$$\sum_{s^{t+1}} \mu(s^{t+1}|s^t) U_{c,t+1} R^b_{t+1} = \sum_{s^{t+1}} \mu(s^{t+1}|s^t) U_{c,t+1} \bar{R}^b_{t+1}, \quad (2.10)$$

$$\sum_{s^{t+1}} \mu(s^{t+1}|s^t) U_{c,t+1} \tau^k_{t+1} (F_{k,t+1} - \delta) = \sum_{s^{t+1}} \mu(s^{t+1}|s^t) U_{c,t+1} \bar{\tau}^k_{t+1} (F_{k,t+1} - \delta) \quad (2.11)$$

$$\tau^k_{t+1} (F_{k,t+1} - \delta) k_{t+1} - R^b_{t+1} b_{t+1} = \bar{\tau}^k_{t+1} (F_{k,t+1} - \delta) k_{t+1} - \bar{R}^b_{t+1} b_{t+1}. \quad (2.12)$$

Degrees of indeterminacy

There are $N-1$ degrees of indeterminacy if, in some period t, s_{t+1} can take on N values. Any set of restrictions across τ^k and R^b that leads to $N-1$ restrictions at each node leads to a unique policy. Examples:

- State-non-contingent capital tax:

$$\tau^k(s^t, s_{t+1}) = \bar{\tau}^k(s^t), \quad \forall s_{t+1}.$$

- State-non-contingent debt return:

$$R^b(s^t, s_{t+1}) = \bar{R}^b(s^t), \quad \forall s_{t+1}.$$

Ex ante capital tax rate

While ex post capital taxes are not pinned down, the **ex ante** tax rate is:

$$\tau_t^e \triangleq \frac{\sum_{s^{t+1}} q_{t+1} \tau_{t+1}^k (F_{k,t+1} - \delta)}{\sum_{s^{t+1}} q_{t+1} (F_{k,t+1} - \delta)}, \tag{2.13}$$

where the state price (SDF) is

$$q_{t+1} \triangleq \beta \mu(s^{t+1}|s^t) \frac{U_{c,t+1}}{U_{c,t}}.$$

Why τ_t^e is determined

Rewrite (2.8) as

$$1-\sum_{s^{t+1}} q_{t+1}=\sum_{s^{t+1}} q_{t+1}(F_{k,t+1}-\delta)-\sum_{s^{t+1}} q_{t+1}\tau_{t+1}^k(F_{k,t+1}-\delta).$$

Thus:

$$\tau_t^e=1-\frac{1-\sum_{s^{t+1}} q_{t+1}}{\sum_{s^{t+1}} q_{t+1}(F_{k,t+1}-\delta)}=\frac{1+\sum_{s^{t+1}} q_{t+1}(F_{k,t+1}-\delta-1)}{\sum_{s^{t+1}} q_{t+1}(F_{k,t+1}-\delta)},$$

which does not depend on the ex post tax rate τ_t^k.

Debt returns and debt taxation

Given (2.12), revenues from capital taxation minus the value of outstanding debt are determined.

- Government can promise a state-non-contingent rate $\bar{r}_t$:

$$\sum_{s^{t+1}} q_{t+1} R^b_{t+1} = \sum_{s^{t+1}} q_{t+1}(1+\bar{r}_t), \tag{2.14}$$

- and levy a state-contingent tax τ^b_{t+1} on interest payments:

$$R^b_{t+1} = 1 + \bar{r}_t(1-\tau^b_{t+1}). \tag{2.15}$$

State-non-contingent rate

From (2.9):

$$\bar{r}_t = \frac{1 - \sum_{s^{t+1}} q_{t+1}}{\sum_{s^{t+1}} q_{t+1}}.$$

From (2.1), b_t is pinned down. Thus (2.12) pins down the sum of tax revenues from capital income tax and debt income tax:

$$\tau^k_{t+1}(F_{k,t+1} - \delta)k_{t+1} + \tau^b_{t+1}\bar{r}_t b_{t+1} = (1 + \bar{r}_t)b_{t+1} + \text{constant}.$$

Average tax rate on assets

Transform revenues into a rate by dividing by income from capital and debt:

$$\tau^a_{t+1} \triangleq \frac{\tau^k_{t+1}(F_{k,t+1}-\delta)k_{t+1} + \tau^b_{t+1}\bar{r}_t b_{t+1}}{(F_{k,t+1}-\delta)k_{t+1} + \bar{r}_t b_{t+1}}. \tag{2.16}$$

- All (certain) tax rates can be described by time-invariant policy rules from period 1 on:

$$\tau^n(k,s;\eta), \quad \tau^e(k,s;\eta), \quad \tau^a(k,s;\eta).$$

- Period-0 policy rules differ from these time-invariant rules.

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Preferences: KPR

King, Plosser, and Rebelo (1988):

$$U(c,n) = \frac{\left(c^{\gamma}(1-n)^{1-\gamma}\right)^{1-\sigma}}{1-\sigma}.$$

Baseline:

$$\sigma = 1 \quad \Longrightarrow \quad U = \gamma \ln c + (1-\gamma)\ln(1-n).$$

Also consider high risk aversion:

$$\sigma = 9.$$

Technology and shocks

Technology:

$$y_t = k_t^{\alpha}[n \cdot \exp(\rho t + z_{t+1})]^{1-\alpha},$$

where

$$z_{t+1} = \rho_z z_t + \varepsilon_t^z, \qquad \varepsilon_t^z \sim \mathcal{N}(0, \sigma_z^2).$$

Government spending:

$$g_t = G \cdot \exp(\rho t + \tilde{g}_t),$$

where

$$\tilde{g}_{t+1} = \rho_g \tilde{g}_t + \varepsilon_t^g, \qquad \varepsilon_t^g \sim \mathcal{N}(0, \sigma_g^2).$$

Calibration targets

Following Christiano and Eichenbaum (1992), use a non-stochastic stationary economy first.

Target	U.S. value
Capital-output ratio	2.71
Fraction of time worked	0.23
Government spending / output	0.18
Debt-output ratio	0.51

Non-stochastic benchmark policies:

$$\tau^k = 27.1\%, \qquad \tau^n = 23.7\%.$$

Initial conditions from deterministic BGP:

$$k_0 = 1.05, \qquad R_0^b b_0 = 0.20.$$

Baseline parameter values

Preferences	$\gamma = .25$	$\sigma = 1$	$\beta = .98$
Technology	$\alpha = .34$	$\delta = .08$	$\rho = .016$
Government spending process	$G = .07$	$\rho_g = .89$	$\sigma_g = .07$
Technology shock process	$\rho_z = .81$	$\sigma_z = .04$	

Numerical procedure: overview

1. De-trend the growing economy.
2. Compute Ramsey allocations for a fixed multiplier η on the IC.
3. Solve for the multiplier η.
4. Compute Ramsey policies.

Step 1: De-trend

- Transform the economy with growth into one without growth.
- Standard procedure.
- This transformation affects only:
  - the discount factor;
  - the depreciation rate.

Step 2: Ramsey allocations for fixed η

Guess η, the multiplier on the implementability constraint. For $t \geq 1$, solutions are stationary functions of (k, s), where

$$s \triangleq (z, \tilde{g}).$$

- Use the resource constraint (1.14) and intratemporal condition (2.3) to express $c(k, s)$ and $n(k, s)$ in terms of k, s, and $k'(k, s)$.
- Use the Euler equation (2.4) to determine $k'(k, s)$.

Step 2: Period 0

For $t = 0$:

- Use the resource constraint (1.14) and intratemporal condition (2.5) to express c_0 and n_0 in terms of k_1.
- Use the Euler equation (2.6) to determine k_1.

Step 2: Bond allocation

For $t \geq 1$, (2.1) implies the bond allocation rule is a stationary function of (k, s) given η. Recursive form:

$$U_c(\bar{s})b(\bar{s}) = \sum_{s'} \beta\mu(s'|s)\left[U_c(\bar{s}')c(\bar{s}')+U_n(\bar{s}')n(\bar{s}')+U_c(\bar{s}')b(\bar{s}')+U_c(\bar{s}')k'(\bar{s}'\right. \tag{3.1}$$

where $\bar{s} \equiv (k, s)$.

- The stationary bond rule is the fixed point of the mapping $b \to b'$ implied by this equation.
- For $t = 0$, substitute period-0 rules into the LHS and stationary rules into the RHS to derive $b_1(\cdot)$.

Computational approximation

- Finding a function $k'(k, s)$ satisfying (2.4) for all k, s is computationally infeasible.
- Restrict to a finite-parameter class of decision rules (technical appendix).
- Similarly, restrict the bond rule to be continuous and piecewise linear in k for each fixed s.
- Require (3.1) to hold at a finite set of points.

Step 3: Solve for the IC multiplier

Substitute $b_1(\cdot)$ and other period-0 allocation rules into the period-0 consumer budget constraint (1.6). Combining (1.7) with (1.8):

$$\frac{U_{c,t}}{U_{n,t}} = -(1-\tau_t^n)w_t.$$

Using this equation and setting

$$\tau_0^k = 27.1\%$$

(s_0 is predetermined), calculate a value for $R_0^b b_0$.

- Iterate on η until

$$R_0^b b_0 = 0.20.$$

Step 4: Compute Ramsey policies

Substitute the allocation rules into:

- Labor income tax:

$$-\frac{U_{n,t}}{U_{c,t}} = (1-\tau_t^n)F_{n,t}.$$

- Ex ante capital income tax:

$$\tau_t^e = \frac{\sum_{s^{t+1}} q_{t+1}\tau_{t+1}^k(F_{k,t+1}-\delta)}{\sum_{s^{t+1}} q_{t+1}(F_{k,t+1}-\delta)}.$$

- Tax on private assets:

$$\tau_{t+1}^a = \frac{\tau_{t+1}^k(F_{k,t+1}-\delta)k_{t+1} + \tau_{t+1}^b \bar{r}_t b_{t+1}}{(F_{k,t+1}-\delta)k_{t+1} + \bar{r}_t b_{t+1}}.$$

Summary

- **Decentralized economy**: linear taxes on labor and capital, short-term debt.
- **Implementability constraint**: summarizes consumer optimality and budget constraint.
- **Ramsey allocations**: maximize preferences subject to IC and resource constraint.
- **Ramsey policies**:
  - labor tax pinned down by intratemporal condition;
  - ex post capital tax and debt returns indeterminate;
  - ex ante capital tax and average asset tax determined.
- **Computation**: de-trend, solve allocations for fixed η, iterate on η, recover policies.

Key takeaway

- The Ramsey problem is not a first-best problem because lump-sum taxes are unavailable.
- The IC captures the distortionary cost of taxation.
- There is a genuine indeterminacy between state-contingent capital taxes and state-contingent debt returns.
- The indeterminate objects do not affect allocations; only certain *ex ante* and *average* tax rates are determined.

1 Decentralized Economy

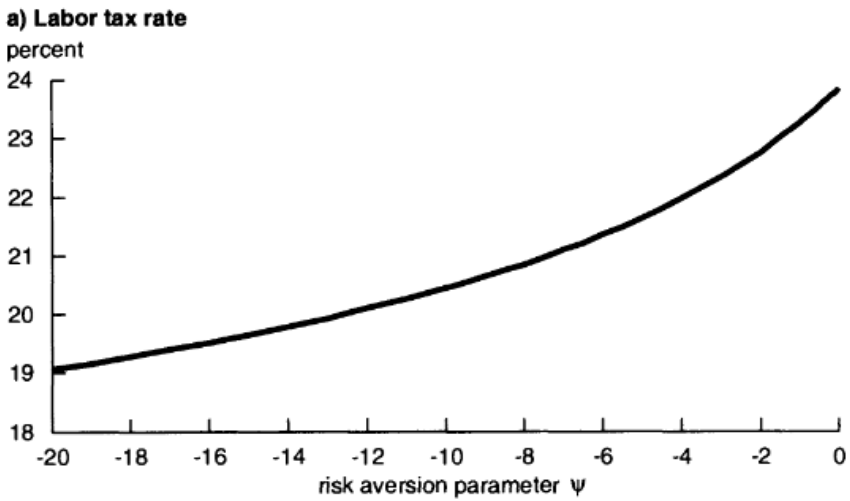
2 Planning Problem

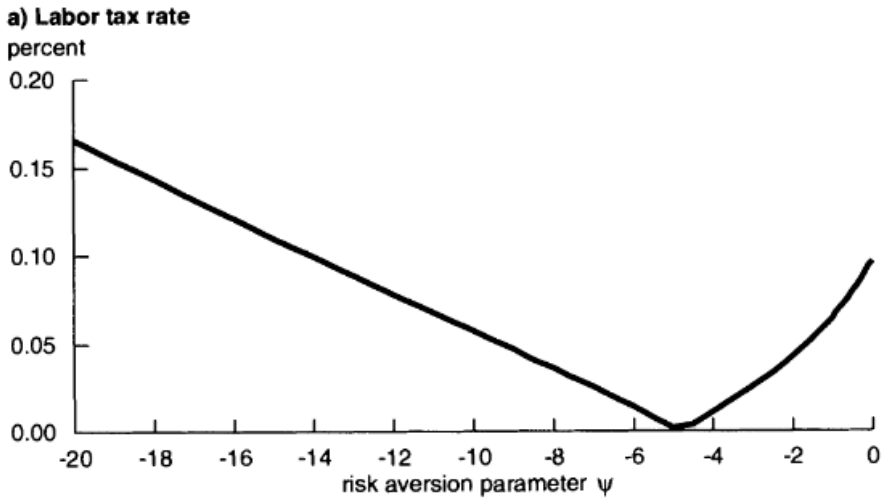
3 Computation Strategy

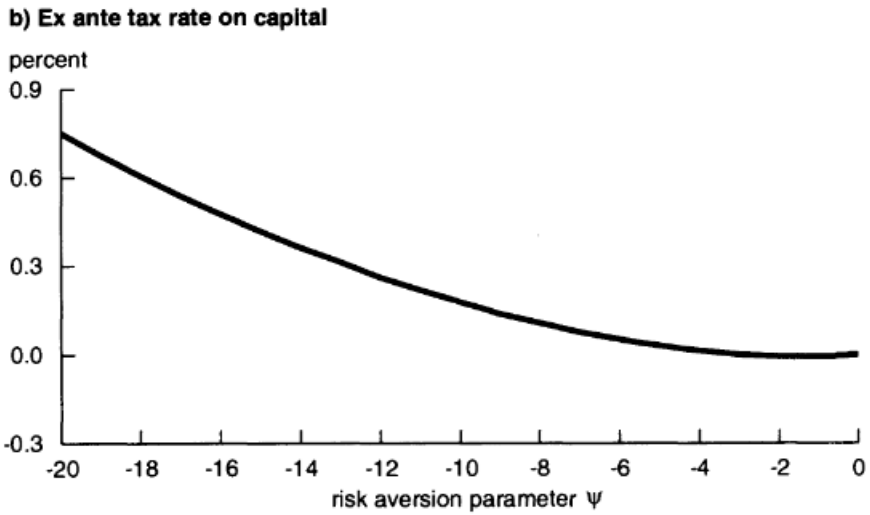
4 Findings

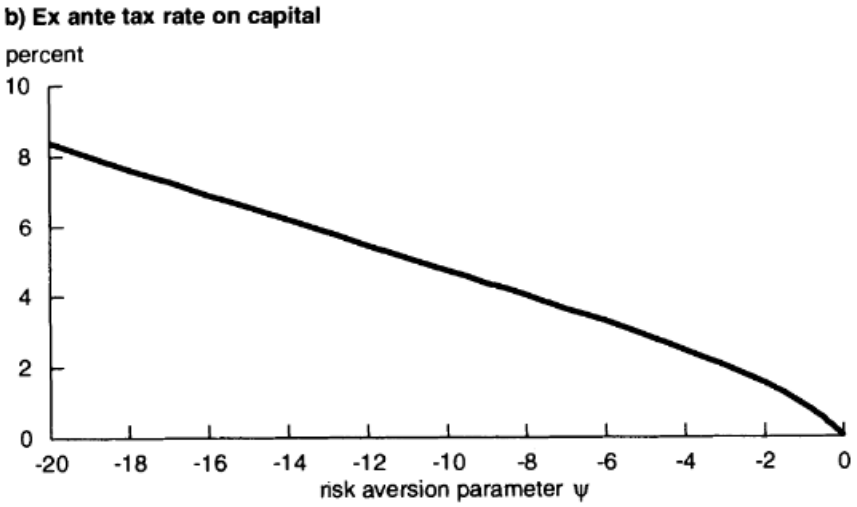
Main Results

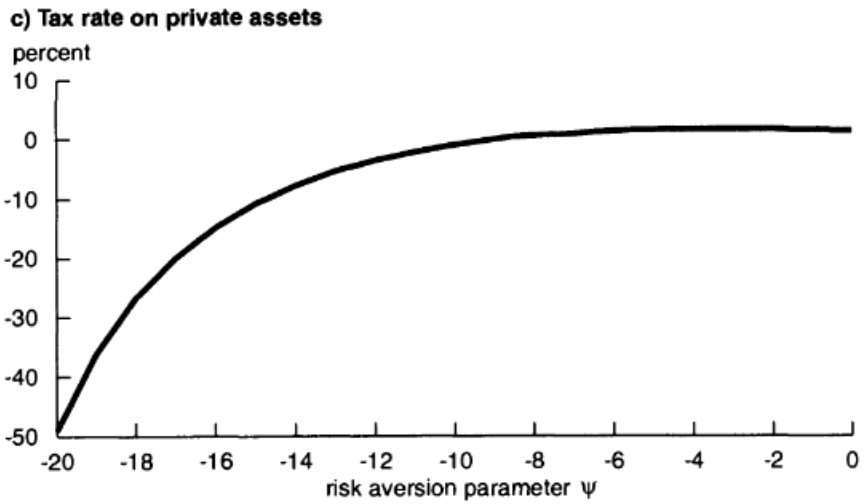
- First, in all the models, the labor tax rate fluctuates very little.
- Second, as is to be expected from proposition 3, in all the models with log utility, the ex ante capital tax rate is identically zero. Both of these findings hold even for the high risk aversion model, which has nonseparable utility: the ex ante capital tax rate is close to zero on average and fluctuates very little.
- Third, the tax rate on private assets is close to zero on average and fluctuates a great deal.

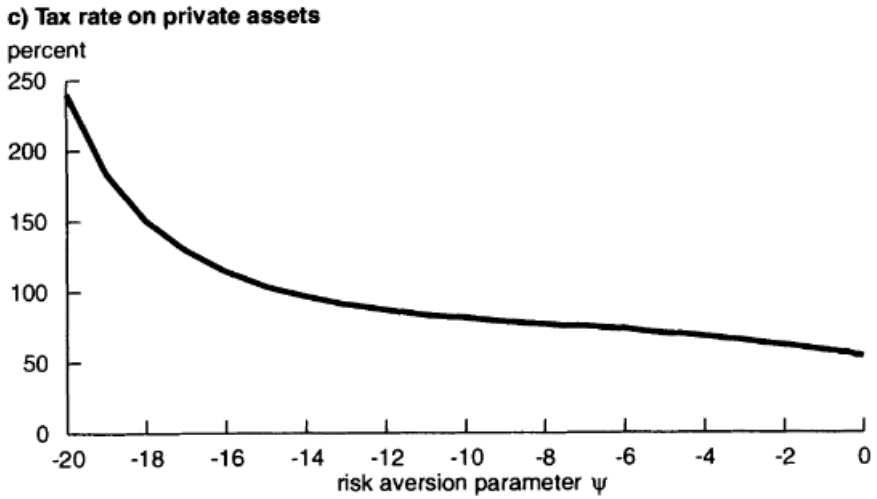

a) Labor tax rate
percent
24
23
22
21
20
19
18
-20
-18
-16
-14
-12
-10
-8
-6
-4
-2
0
risk aversion parameter ψ

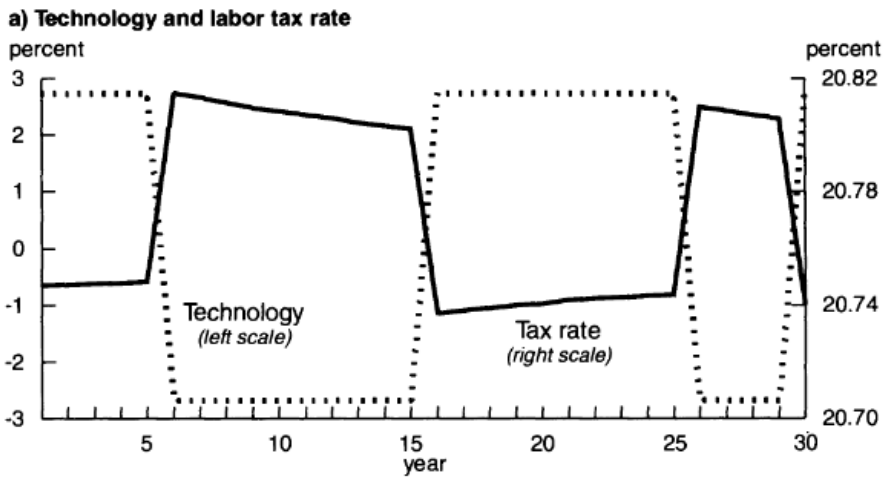

a) Labor tax rate
percent
0.20
0.15
0.10
0.05
0.00
-20
-18
-16
-14
-12
-10
-8
-6
-4
-2
0
risk aversion parameter ψ

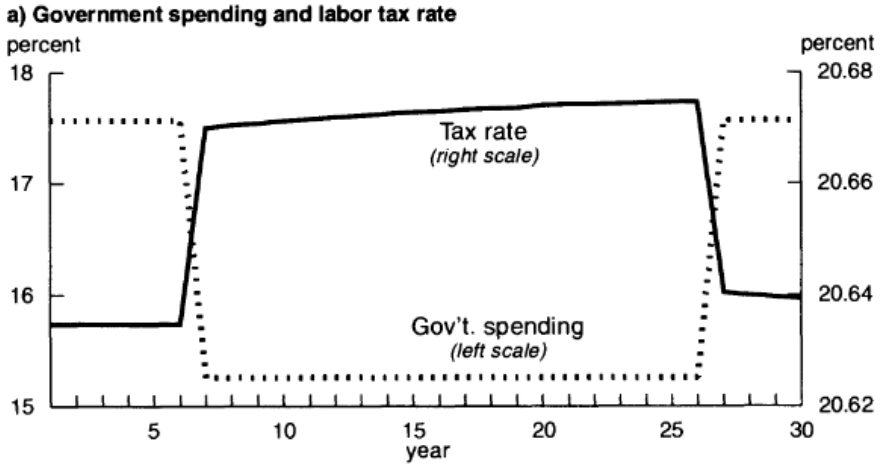

b) Ex ante tax rate on capital
percent
0.9
0.6
0.3
0.0
-0.3
-20
-18
-16
-14
-12
-10
-8
-6
-4
-2
0
risk aversion parameter ψ


b) Ex ante tax rate on capital
percent
10
8
6
4
2
0
-20
-18
-16
-14
-12
-10
-8
-6
-4
-2
0
risk aversion parameter ψ

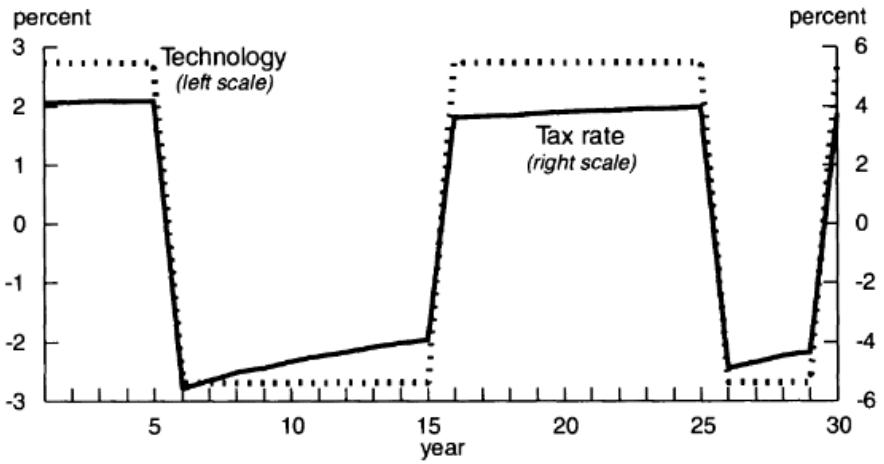

c) Tax rate on private assets
percent
10
0
-10
-20
-30
-40
-50
-20
-18
-16
-14
-12
-10
-8
-6
-4
-2
0
risk aversion parameter ψ


c) Tax rate on private assets
percent
250
200
150
100
50
0
-20
-18
-16
-14
-12
-10
-8
-6
-4
-2
0
risk aversion parameter ψ

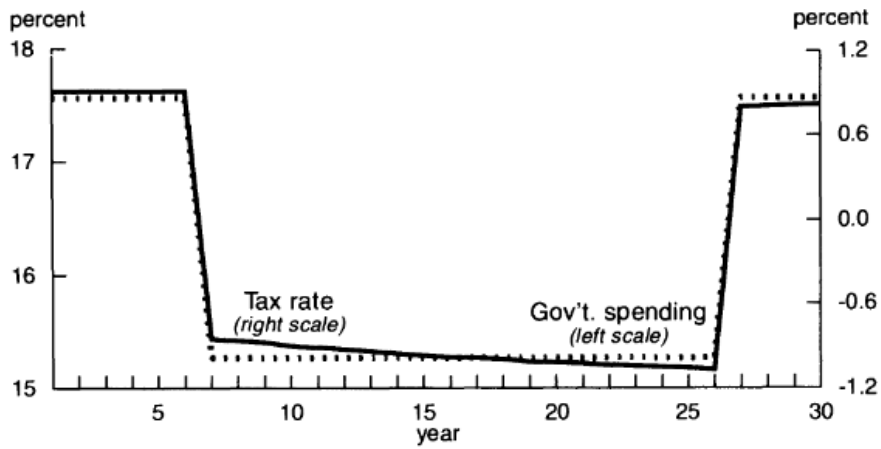

a) Technology and labor tax rate
percent
percent
3
2
1
0
-1
-2
-3
20.82
20.78
20.74
20.70
Technology
(left scale)
Tax rate
(right scale)
5
10
15
20
25
30
year


a) Government spending and labor tax rate
percent
percent
18
17
16
15
20.68
20.66
20.64
20.62
Tax rate
(right scale)
Gov't. spending
(left scale)
5
10
15
20
25
30
year

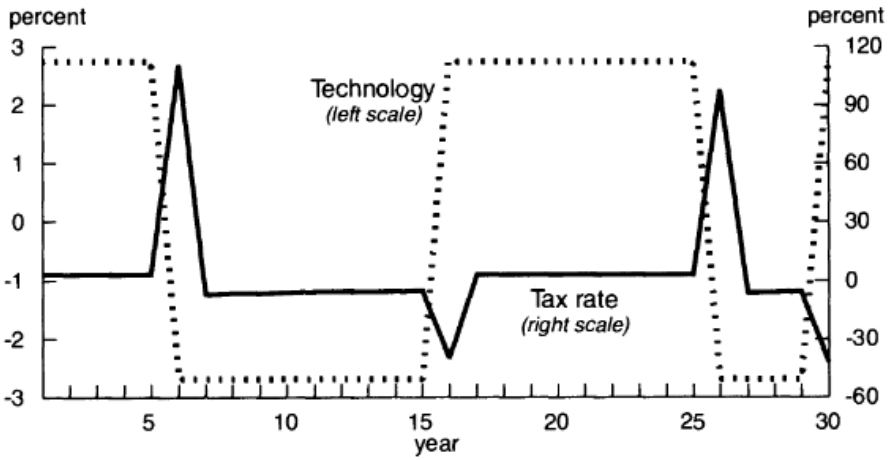
b) Technology and ex ante tax rate on capital



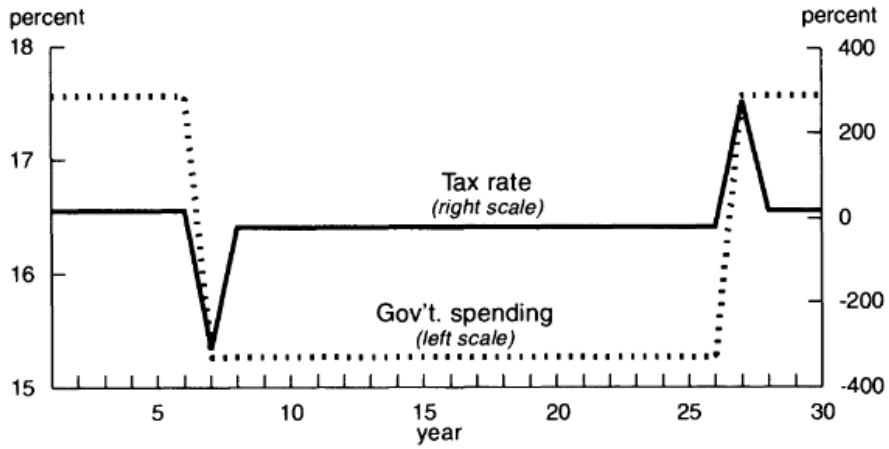
b) Government spending and ex ante tax rate on capital



c) Technology and tax rate on private assets



c) Government spending and tax rate on private assets



Benchmark Economies	Ramsey	Policy Experiments: Constant Tax Rates	
		Zero Capital Tax	High Capital Tax
		Log Utility	
Deterministic economy	1.00	.20	.00
Stochastic economy:			
Estimated policy	1.00	.20	.03
Variable policy	1.60	.80	.60
		High Risk Aversion	
Deterministic economy	1.30	−.06	.00
Stochastic economy:			
Estimated policy	1.20	−.02	.03
Variable policy	1.60	.33	.38
		High Initial Debt	
Deterministic economy	5.20	−.70	.00
Stochastic economy:			
Estimated policy	5.10	−.60	.05
Variable policy	6.10	.30	1.00

Questions are welcome.